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SUPERSEDING
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MILITARY SPECIFICATION

STAMPINGS, STEEL, SEAMLESS, COLD-DRAWN;
FOR MANUFACTURE OF SHELL FOR SMOOTH-BORE MORTARS

1. SCOPE

1.1 This specification covers cold-drawn, seamless steel stampings, formed from plate, for manufacture of shell for smooth-bore mortars.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, and the Ordnance Corps drawing pertaining to the stamping under contract, and all drawings and specifications referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-M-151	-Metals; General Specification For Inspection Of

MILITARY

MIL-S-13852	-Steel, Disks for Deep Drawn Ammunition Items
MIL-C-490	-Cleaning and Preparation of Ferrous and Zinc-Coated Surfaces For Organic Protective Coatings
MIL-G-2550	-General Specification For Ammunition Except Small Arms Ammunition
MIL-P-10025	-Packing And Marking For Domestic Shipment Of Inert Ammunition Components; General Specification for

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.-Unless otherwise specified, stampings shall be made from steel complying with Specification MIL-S-13852.

3.2 The stampings shall comply with all requirements on the applicable drawing, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Stampings shall be so formed that the shell cavity is formed to finish dimensions without machining, except for nose threads and base plug, if used. Wherever machining is required to produce a finished shell, the parts of the stampings that require such machining shall be machinable, and shall be free from hard spots and other discontinuities.

3.4 When base plugs are silver-brazed in place, the silver brazing shall be done by electric induction heating, or by a method approved by the responsible procurement agency. The heating so performed shall be localized to the minimum area consistent with good brazing practice. The technique employed in the silver-brazing process shall be uniformly maintained so that the brazed bonds on all stampings comply with the requirements specified on the applicable drawings and in the applicable specifications, when examined as specified in 4.4.

3.5 Stress relief anneal.-Stampings shall be stress-relieved as soon as practicable after drawing. Unless otherwise approved by the responsible procurement agency, all stampings shall be brought to a uniform temperature of not less than 700°F, and maintained at that temperature for not less than 1 hour. On stampings incorporating a brazed plug in the base end, stress relieving shall be performed prior to assembly of the plug.

3.6 Stampings incorporating base plug silver-brazed in place.

3.6.1 The base plug of all silver-brazed stampings shall withstand the impact test specified on the stamping drawing, when tested as specified in 4.5.

3.6.2 The stampings with silver-brazed plugs shall withstand an internal air pressure of 150 pounds per square inch, minimum, for a minimum of 15 seconds, without showing evidence of leaks, when tested as specified in 4.6.

3.6.3 All stampings shall withstand the static load test applied on the plug, as specified on the stamping drawing or on the shell drawing, without failure of the bond produced by silver brazing, when tested as specified in 4.8.

3.7 Mechanical properties.-Unless otherwise approved by the responsible procurement agency, stampings shall comply with the mechanical requirements (see 6.2) specified for the shell for which the stamping is intended. Unless otherwise specified on the shell drawing, or on the stamping drawing, stampings shall comply with the following mechanical requirements, when tested as specified in 4.9.

Yield strength, minimum	35,000 psi
Elongation in 2 inches, minimum	15 percent

3.8 Cleaning of stampings

3.8.1 The entire cavity of each stamping shall be thoroughly cleaned by the shot or sand blast method, or by a method approved by the responsible procurement agency, remove all flux, scale, rust, and other foreign material.

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After cleaning, the stamping cavity shall comply with the requirements of Specification MIL-C-490, except the freedom from oil test, when tested as specified in 4.10.

3.8.2 Immediately after cleaning, the entire cavity of each stamping shall be protected against corrosion by a suitable coating which will withstand a 12-hour salt spray test without the appearance of rust, when tested as specified in 4.10. The protective coating shall be eaily removable by washing or degreasing. The same coating used in the cavity may also be applied to the exterior surfaces of the stamping if desired.

3.8.3 Application of the protective coating, **(see 4.10) to** the cavity is optional when the stamping is machined at the same plant, by the same contractor, and the stamping is processed through the machining line immediately after cleaning.

3.9 Workmanship.-Stampings shall be sound and free from cracks, porosity, and other similar discontinuities. The interior of stampings shall be smooth, and free from burrs, fins, deep scars, pits, sharp edges, **scale,** and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.

4.1.1 Production lot.-A lot shall consist of all stampings made by one manufacturer under one contract, in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Any number of heats or melts of steel may be represented in a lot of stampings. However, the number of heats or melts of steel represented in a lot shall be kept to a minimum consistent with good production practices. All heats or melts represented in any one lot of stampings shall be indicated on the Ammunition Data Cards. A lot of stampings shall consist only of stampings made from blanked disks of the same nominal thickness and the same nominal diameter.

4.1.2 Pilot lot.-A pilot lot shall consist of five stampings made from material of one manufacturer by the same process intended for production in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

4.2 Inspection.-Inspection shall be as specified in Specifications QQ-M-151 and MIL-G-2550.

4.3 Pilot lot test.-The five sample stampings submitted for pilot lot tests shall be forwarded, as directed by the responsible procurement agency, to a Government arsenal for metallurgical examination, and for tests of the strength of the silver-brazed bond if the stampings incorporate base plugs.

4.4 Examination of silver-brazed bond.

4.4.1 Three stampings shall be sectioned longitudinally, and the two half-sections of the plug removed by any suitable method not involving the application of heat. The mating surfaces between the plug and stamping shall show evidence of a minimum bonded area of 75 percent, with no unbonded region extending the full length of the plug.

4.4.2 Acceptance.-The lot shall be considered acceptable provided the three stampings show evidence of a minimum bonded area of 75 percent, with

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no unbonded region extending the full length of the plug. If any stamping fails to comply with this requirement, the lot shall be considered eligible for retest under the provisions of 4.4.3.

4.4.3 Retest.-A retest as provided for in 4.4.2 may be made using ten additional stampings. The lot shall be considered acceptable on retest provided the ten stampings show evidence of a minimum bonded area of 75 percent, with no unbonded region extending the full length of the plug.

4.5 Impact test.-All stampings with base plugs silver-brazed in place shall be subjected to this test. The specified impact load shall be applied to the silver-brazed plug so that the shock is carried by the brazed bond in a longitudinal direction. The impact shall leave a positive identifying mark on the base end of the plug for subsequent evidence that the operation has been performed, to determine compliance with 3.6.1. Any measurable movement of the plug with respect to the body of the stamping will be cause for rejection, subject to repair and retest (see 4.7.)

4.6 Air pressure test.-Subsequent to the impact test, all stampings with base plugs silver-brazed in place shall be subjected to an internal air pressure of 150 pounds per square inch, minimum, for not less than 15 seconds, to determine compliance with 3.6.2. The exterior of the joint between the stamping and the base plug shall be submerged in water in such a manner as to be clearly visible during the test. All stampings which show leaks shall be rejected subject to repair and retest (see 4.7).

4.7 Repair and retest.-Stampings rejected in the impact test (see 4.5) or in the air pressure test (see 4.6) because of failure to obtain a suitable silver-brazed bond may be repaired and resubmitted for test, but the method of repair shall involve removal of the plug and subjection to the complete brazing operation once again. Repair of any portion of the stamping by welding is not authorized.

4.8 Static load test.

4.8.1 Ten samples shall be selected by the inspector and static load shall be applied in such a manner that the load is absorbed in shear stress of the silver-brazed bond, to determine compliance with 3.6.3.

4.8.2 Acceptance.-The lot shall be considered acceptable provided the ten stampings comply with the requirement of 3.6.3. If any stampings fails to comply with this requirement, the lot shall be considered eligible for retest under the provisions of 4.4.3.

4.8.3 Retest.-A retest as provided for in 4.8.2 may be made using 20 additional stampings. The lot shall be acceptable on retest provided all the stampings comply with the requirement of 3.6.3.

4.9 Tension tests.

4.9.1 Preparation of test specimens.-Two test specimens shall be cut from each stamping selected for test. The test specimens shall be taken longitudinally from diametrically opposite sides of the wall of each of the sample stampings. The end of the specimens shall be located approximately in line with the bottom of the cavity, except where the design of stamping is such that there is a discontinuity in body contour between the body and base portions, in which case an equal number of test specimens shall be taken from both the center portion of the body and the base portion of the stampings.

Tension test specimens shall conform to Type 1 or the largest size of Type 4 obtainable (see Specification QQ-M-151) or, if the wall thickness of

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the stamping is not sufficient for Type 1 or 4, to the largest size of Type 5 or Type 5a specimen obtainable. For thin-walled shell, the contractor may use flat machined test specimens in place of Type 4 specimens to obtain larger cross-section area for determining tensile properties. Such specimens shall be machined to the largest size obtainable, and shall have a width equal to at least twice the thickness, and a gage length equal to at least four times the width. However, in case of dispute arising from the use of such specimens, final decision shall be based upon results of test specimens complying with the requirements of Specification QQ-M-151.

4.9.2 Yield strength shall be determined by either the offset method or the extension-under-load method, as specified in Specification QQ-M-151. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length). The limiting extension-under-load shall be calculated as follows, to determine compliance with 3.7:

$$X = \frac{Y}{30{,}000{,}000} + 0.001$$

where:

X = Limiting extension, inch per inch of gage length.
Y = Specified yield strength.

4.9.3 Acceptance.-The lot shall be considered acceptable provided the test specimens representing a lot of stampings comply with the mechanical requirements specified in 3.7. If any test specimen fails to comply with the mechanical requirements, the lot shall be considered eligible for retest under the provisions of 4.9.4.

4.9.4 **Retest**.-A retest as provided for in 4.9.3 may be made using double the number of test specimens as were used in the original test. The lot shall be considered acceptable on retest provided all the specimens comply with the physical requirements of 3.7.

4.10 Cleanliness and protective coating.-The inspector shall assure himself that the operations of cleaning the cavity of the stamping are performed in such a manner as to remove all foreign material, and insure that the surfaces comply with the requirements of Specification MIL-C-490 and that the protective coating is applied to the cavity surfaces after cleaning so that the surfaces withstand the salt spray test conducted in accordance with Specification QQ-M-151, to determine compliance with 3.8.

5. PREPARATION FOR DELIVERY

5.1 Packing.-All shipments of stampings shall be made in closed box cars, closed-body trucks, or other closed and protected carrier, which are clean and structurally sound. All shipments shall be loaded in such a manner as to insure acceptance, by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery.

5.2 Marking.-Shipments shall be marked in accordance with Specification MIL-P-10025.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

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6. NOTES

6.1 Ordering data.-Purchasing documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity and dimensions.
c. Applicable drawing numbers.
d. Material, if other provided in 3.1.
e. Mechanical property requirements, if other than provided in 3.7.

6.2 Mechanical properties should not be determined by hardness tests in place of tensile test coupons, except where specifically authorized on the stamping drawing in regions of the stamping where tensile test coupons cannot be conveniently obtained. When permitted, the hardness survey should be made over the length of the stamping, and include the hardness of the regions where the tensile test coupons are obtained. Results of hardness surveys are comparative only within any one stamping, and should not be used to comply mechanical properties between stampings. Hardness is not proportional to yield strength in cold-worked material (see 3.7.)

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